

Site24x7 Disk Space Report

Full Automation Suite — Technical Documentation

Site24x7 · InfluxDB · Grafana · Outlook Email

Organisation: [Organisation]

Team: DevOps

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Version: 2.0

Status: Active

1. Executive Summary

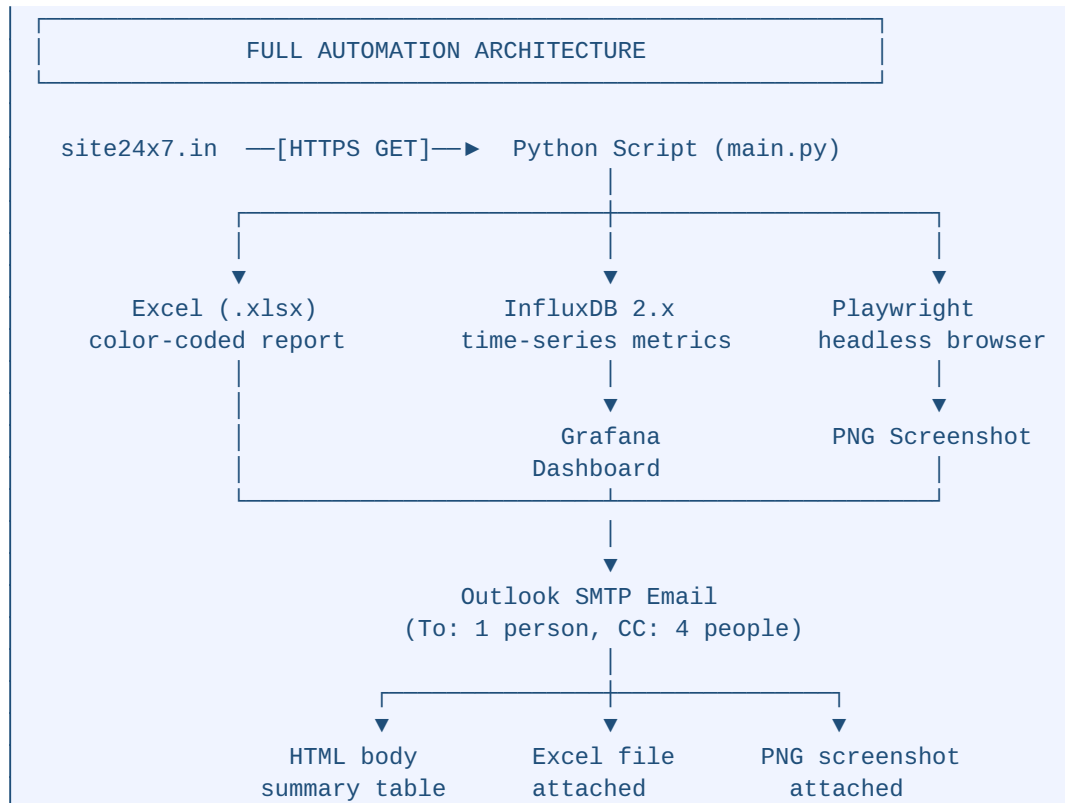
This document is the complete technical reference for the Site24x7 Disk Space Report Automation Suite developed by the DevOps team at [Organisation] . It covers design, implementation, authentication validation, configuration, error handling, and future roadmap.

The suite replaces a fully manual daily process — logging into Site24x7, copying disk metrics server by server, pasting into Excel, and emailing the sheet — with a single Python script that runs automatically every morning at 09:00 AM.

Component	Purpose	Status
Site24x7 REST API	Source of all disk and server health data	Connected
Python Script	Orchestrates all steps in a single run	Active
InfluxDB 2.x	Time-series storage for Grafana visualisation	Setup required
Grafana	Live dashboard with historical trending and alerts	Setup required
Playwright	Headless browser for Grafana dashboard screenshot	Active
Outlook / SMTP	Automated email delivery with attachments	Active

2. Architecture and Data Flow

2.1 End-to-End Flow



2.2 Script Execution Steps

Step	Function	Action	Skip Condition
1	<code>fetch_disk_data()</code>	Authenticate + pull server metrics from Site24x7 API	Token not set → exits
2	<code>generate_excel()</code>	Build color-coded .xlsx with status and disk thresholds	Never skipped
3	<code>push_to_influxdb()</code>	Write disk metrics to InfluxDB for Grafana	Token not set → skips gracefully
4	<code>screenshot_grafana()</code>	Login to Grafana, open dashboard, capture PNG	URL not configured → skips
5	<code>send_email()</code>	Send HTML email with table, Excel, PNG via Outlook SMTP	Email not configured → skips

3. Authentication — Setup and Validation

This section covers how to obtain credentials for every service and how to verify each one is working correctly **before** running the full script. Always validate authentication first — a bad token causes confusing errors downstream.

3.1 Site24x7 — Zoho Authtoken

How to Get the Token

- Log in to <https://www.site24x7.in> in your browser.
- Open a new tab and go to:

```
| https://accounts.zoho.in/apiauthtoken/create?SCOPE=Site24x7/site24x7api
```

- The page returns plain text. Copy the value after AUTHTOKEN=

```
AUTHTOKEN=YOUR_TOKEN_HERE  
RESULT=TRUE
```

← copy this part only

- Paste this value into main.py at SITE24X7_TOKEN.

How to Validate the Token

Run this command in terminal to test the token before running the full script:

```
# Windows PowerShell:  
curl -H "Authorization: Zoho-authtoken YOUR_TOKEN_HERE" \  
-H "Accept: application/json; version=2.0" \  
https://www.site24x7.in/api/current_status  
  
# Expected: JSON response starting with {"code":0, ...}  
# Bad token: {"code":1001, "message":"Invalid authtoken"}
```

Token Validation Checklist

Check	Expected Result	If Wrong
RESULT=TRUE on token page	Page shows RESULT=TRUE	Account may not have API permission — ask admin
curl returns code:0	JSON with code:0 in response	Token is wrong or expired — regenerate
curl returns 200 HTTP status	No HTTP error	Network issue or wrong data center URL

Check	Expected Result	If Wrong
monitors array is not empty	List of your monitors in response	Account has no monitors configured

Token Expiry

Zoho authtokens do not expire automatically but can be revoked. If the script suddenly fails with a 401 error, regenerate the token from the same URL above and update main.py.

3.2 InfluxDB — API Token

How to Get the Token

- Install and start InfluxDB — see Section 4.1 for full installation steps.
- Open `http://localhost:8086` in browser.
- On first run click Get Started and fill in:
 - Username: admin
 - Password: choose a strong password
 - Organisation: `your-org`
 - Bucket: disk-metrics
- Click Continue — the next screen shows your API token. Copy it immediately.
- If you missed it: go to Data → API Tokens → Generate API Token → All Access.

How to Validate the Token

```
# Run in terminal – should return HTTP 200 and empty JSON {}
curl -H "Authorization: Token YOUR_INFLUX_TOKEN" \
      http://localhost:8086/api/v2/buckets?org=your-org

# Expected: JSON list of buckets including disk-metrics
# Bad token: {"code":"unauthorized","message":"unauthorized access"}
```

InfluxDB Auth Checklist

Check	Expected	If Wrong
InfluxDB is running	<code>http://localhost:8086</code> loads in browser	Start InfluxDB: run <code>influxd.exe</code>
Organisation name matches	<code>INFLUX_ORG</code> in script = org in InfluxDB	Update <code>INFLUX_ORG</code> in <code>main.py</code>
Bucket exists	disk-metrics bucket visible in UI	Create bucket in InfluxDB UI → Data → Buckets
Token has write access	curl returns bucket list	Regenerate as All Access token

3.3 Grafana — Username and Password

Default Credentials

```
Username: admin
Password: admin (you will be prompted to change on first login)
```

How to Validate Grafana Access

```
# Test Grafana API – should return {"message":"Logged in"}
curl -u admin:YOUR_GRAFANA_PASSWORD \
      http://localhost:3000/api/login/ping

# Test dashboard exists – replace DASHBOARD_UID with yours
curl -u admin:YOUR_GRAFANA_PASSWORD \
      http://localhost:3000/api/dashboards/uid/DASHBOARD_UID
```

Getting Your Dashboard URL

- Open your dashboard in Grafana browser.
- Copy the full URL from address bar — example:

```
http://localhost:3000/d/abc123xyz/disk-space-report
```

- Paste this exact URL into `GRAFANA_DASHBOARD_URL` in `main.py`.
- The screenshot step opens this URL in a headless browser — if URL is wrong, screenshot will be blank.

Grafana Auth Checklist

Check	Expected	If Wrong
Grafana running	http://localhost:3000 loads	Start Grafana service
Login works	curl /api/login/ping returns 200	Wrong password — reset in Grafana UI
InfluxDB datasource added	Data Sources page shows InfluxDB	Add datasource — Section 4.3
Dashboard URL is correct	Dashboard loads in browser from that URL	Copy URL again from browser address bar
Panels show data	Gauge/chart shows disk % values	Check Flux query — Section 4.4

3.4 Outlook / Office 365 — SMTP Authentication

Standard Login (No MFA)

If your company Outlook account does not have multi-factor authentication enabled:

- EMAIL_FROM = your full Outlook email address
- EMAIL_PASSWORD = your regular Outlook password

App Password (If MFA is Enabled)

Important: If your account has 2FA or MFA enabled, your regular password will NOT work with SMTP. You must create an App Password.

- Go to <https://account.microsoft.com/security>
- Click Advanced Security Options
- Scroll to App Passwords → click Create a new app password
- Name it: Site24x7 Automation
- Copy the generated password (shown only once)
- Paste into EMAIL_PASSWORD in main.py — use this instead of your regular password

How to Validate Email Authentication

```
# Run this small Python test script separately before running main.py
import smtplib, ssl

EMAIL = "your.email@company.com"
PASS = "your_password_or_app_password"

try:
    with smtplib.SMTP("smtp.office365.com", 587) as s:
        s.starttls(ssl.create_default_context())
        s.login(EMAIL, PASS)
        print("SUCCESS – Email authentication works")
except Exception as e:
    print(f"FAILED – {e}")
```

Email Auth Checklist

Check	Expected	If Wrong
SMTP test script runs	Prints: SUCCESS — Email authentication works	Wrong password or App Password needed
EMAIL_FROM is full address	your.name@company.com — not just username	Add @company.com to EMAIL_FROM
MFA check	If MFA enabled — App Password must be used	Generate App Password from Microsoft account
CC list format	List of strings in square brackets	Each email in quotes separated by commas

Check	Expected	If Wrong
SMTP port is 587	Port 587 with STARTTLS — do not change	Check with IT if company blocks port 587

4. Installation Guide

4.1 Install InfluxDB

- Go to <https://portal.influxdata.com/downloads/>
- Select InfluxDB 2.x → Windows → download .zip
- Extract the zip to a folder e.g. C:\InfluxDB
- Open Command Prompt in that folder and run:

```
| influxd.exe
```

- Open browser → <http://localhost:8086> → click Get Started
- Fill in: Username, Password, Organisation (your-org), Bucket (disk-metrics)
- Copy and save the API token shown on the next screen

4.2 Install Grafana

- Go to <https://grafana.com/grafana/download?platform=windows>
- Download and run the Windows installer
- Grafana starts as a Windows service automatically
- Open browser → <http://localhost:3000>
- Login: admin / admin → change password when prompted

4.3 Connect InfluxDB to Grafana

- Grafana → Configuration (gear icon) → Data Sources
- Click Add data source → select InfluxDB
- Fill in:
 - Query Language: Flux
 - URL: <http://localhost:8086>
 - Organisation: your-org
 - Token: paste your InfluxDB API token
 - Default Bucket: disk-metrics
- Click Save & Test — should show "datasource is working"

4.4 Create Grafana Dashboard

- Grafana → Dashboards → New Dashboard → Add new panel
- Paste this Flux query in the query editor:

```
| from(bucket: "disk-metrics")
```

```
|> range(start: -24h)
|> filter(fn: (r) => r._measurement == "disk_usage")
|> filter(fn: (r) => r._field == "used_percent")
|> last()
```

- Set visualisation type to Gauge
- Set thresholds: 80 = yellow, 90 = red
- Click Apply → Save dashboard → name it Disk Space Report
- Copy the dashboard URL from browser → paste into GRAFANA_DASHBOARD_URL in main.py

4.5 Install Python Libraries

```
pip install requests openpyxl influxdb-client playwright schedule
playwright install chromium
```

5. Configuration Reference

Open main.py and fill in the CONFIGURATION section at the top. All settings are in one place — nothing else needs to be changed.

Field	Example Value	Where to Get It
SITE24X7_TOKEN	ba4604e8e433g9c892e...	accounts.zoho.in/apiauthtoken/create
TARGET_SERVERS	["Server-01", "Server-02"]	Exact names from Site24x7 monitor list
INFLUX_URL	http://localhost:8086	Default if InfluxDB on same machine
INFLUX_TOKEN	abc123xyz...	InfluxDB UI → Data → API Tokens
INFLUX_ORG	your-org	Organisation name chosen during InfluxDB setup
INFLUX_BUCKET	disk-metrics	Bucket name chosen during InfluxDB setup
GRAFANA_URL	http://localhost:3000	Default if Grafana on same machine
GRAFANA_DASHBOARD_URL	http://localhost:3000/d/abc123/disk	Copy from browser address bar
GRAFANA_USERNAME	admin	Grafana login username
GRAFANA_PASSWORD	your_grafana_password	Password set during Grafana first login
EMAIL_FROM	user@company.com	Your full Outlook email address
EMAIL_PASSWORD	app_password_here	Outlook password or App Password if MFA
EMAIL_TO	manager@company.com	Main recipient — Send To
EMAIL_CC	["p1@co.com", "p2@co.com"]	List of CC recipients — max 4

6. Running the Script

6.1 Recommended Run Order (First Time)

Order	Action	Purpose
1	Run email validation test script (Section 3.4)	Confirm SMTP auth works before anything else
2	Run Site24x7 curl validation (Section 3.1)	Confirm token and server data is accessible
3	Confirm InfluxDB running and bucket exists	Confirm push target is ready
4	Confirm Grafana dashboard URL loads in browser	Confirm screenshot target is correct
5	python main.py	Full run — all 5 steps

6.2 Run Commands

```
# Run once – all 5 steps immediately
python main.py

# Run on schedule – auto-runs daily at 09:00 AM
# Keep this terminal open or run in background
python main.py --schedule

# Run in background on Windows (PowerShell):
Start-Process python -ArgumentList "main.py --schedule" -WindowStyle Hidden
```

6.3 Expected Terminal Output

```
=====
Site24x7 Full Automation Suite
10 August 2026, 09:00:01
=====

[1/5] Fetching data from Site24x7...
[OK]   Matched 3 servers.

[2/5] Generating Excel report...
[OK]   Saved: Disk_Report_2026-08-10.xlsx

[3/5] Pushing to InfluxDB...
[OK]   Pushed 3 records to InfluxDB.

[4/5] Screenshotting Grafana dashboard...
[OK]   Screenshot: Grafana_2026-08-10.png

[5/5] Sending email...
[OK]   Attached: Disk_Report_2026-08-10.xlsx
[OK]   Attached: Grafana_2026-08-10.png
[OK]   Email sent to manager@company.com + 4 CC.
```


7. Output Documentation

7.1 Excel Report — Disk_Report_YYYY-MM-DD.xlsx

Column	Content	Source
Monitor Name	Server monitor name	Site24x7 API
Hostname	Server hostname	Site24x7 API
Status	UP / DOWN / TROUBLE / SUSPENDED	Site24x7 status code
Free Disk Space	Available space with unit (GB/MB)	Site24x7 disk_free attribute
Used Disk %	Percentage disk used	Site24x7 disk_used attribute
Last Polled	When Site24x7 last checked the server	Site24x7 API
Report Time	Script run timestamp	Python datetime.now()

7.2 Excel Color Coding

Color	Column	Condition	Meaning
Green	Status	Status = UP	Server is healthy
Red	Status	Status = DOWN	Server is down — action required
Orange	Status	Status = TROUBLE	Server has issues — investigate
Yellow	Used Disk %	Used >= 80%	Warning — disk filling up
Red	Used Disk %	Used >= 90%	Critical — disk almost full
Light blue	All columns	Alternate rows	Readability shading

7.3 Email Contents

Element	Description
Subject	Daily Disk Space Report — 10 August 2026
To	One main recipient — manager or team lead
CC	Up to 4 team members
Body — Summary boxes	Count of UP / DOWN / TROUBLE / TOTAL servers as colored boxes
Body — Data table	Full disk metrics table with color coding — same as Excel
Body — Legend	Explains yellow and red threshold colors
Body — Button	Blue button linking to live Grafana dashboard (if configured)
Attachment 1	Disk_Report_YYYY-MM-DD.xlsx — color-coded Excel file
Attachment 2	Grafana_YYYY-MM-DD.png — screenshot of live dashboard
Footer	Automated report notice — DevOps team attribution

8. Error Handling Reference

Error Message	Cause	Fix
[ERROR] Set SITE24X7_TOKEN	Token field still has placeholder text	Paste real token into SITE24X7_TOKEN
[ERROR] Invalid token (401)	Token wrong, expired, or revoked	Regenerate from accounts.zoho.in
[ERROR] Cannot connect to site24x7.in	No internet or VPN not connected	Check network connection
[WARN] Server not found	Server name typo in TARGET_SERVERS	Check exact name in Site24x7 monitor list
[SKIP] InfluxDB token not set	INFLUX_TOKEN still placeholder	Set up InfluxDB and paste token
[WARN] InfluxDB connection refused	InfluxDB not running	Start influxd.exe
[SKIP] Grafana not configured	GRAFANA_DASHBOARD_URL not set	Paste dashboard URL from browser
[WARN] Screenshot failed	Wrong URL, Grafana down, or wrong password	Validate Grafana with curl test in Section 3.3
[SKIP] Email not configured	EMAIL_FROM still placeholder	Fill in email config fields
SMTPAuthenticationError	Wrong password or MFA blocking SMTP	Use App Password — see Section 3.4
SMTPConnectError	Port 587 blocked by company firewall	Ask IT to whitelist smtp.office365.com:587

9. Future Implementation Roadmap

9.1 Short Term — 1 to 3 Months

Feature	Description	Priority
Environment variables	Move all tokens/passwords out of script into .env file	High
Multiple drives/partitions	Report disk usage per drive letter (C:, D:, E:) per server	High
Threshold SMS alert	Send SMS via Twilio if any server crosses 90%	High
Config file	Move TARGET_SERVERS and settings to external config.json	Medium
Retry logic	Auto-retry API calls on timeout instead of exiting	Medium
Historical Excel master	Append daily data to one master sheet for trend tracking	Medium

9.2 Medium Term — 3 to 6 Months

Feature	Description	Priority
Grafana alerts	Grafana sends alert if disk > 80% without Python needed	High
CPU and RAM metrics	Extend to include CPU %, memory usage, network I/O	High
Multi-sheet Excel	Separate sheet per server group or team	Medium
Flask web dashboard	Simple internal web page to view latest report in browser	Medium
Slack notifications	Post disk summary to team Slack channel daily	Medium
Anomaly detection	Alert when disk usage grows > 5% day-over-day	Low

9.3 Long Term — 6 to 12 Months

Feature	Description	Priority
Prometheus exporter	Replace Site24x7 with Prometheus node_exporter directly	High
Kubernetes CronJob	Run script as K8s CronJob — no machine dependency	Medium
AWS S3 archival	Store daily reports to S3 for long-term access	Medium
REST API	Expose disk metrics as internal REST endpoint for other tools	Low
Full Grafana migration	Script becomes redundant — Grafana + Prometheus handles all	High

10. Security Considerations

Area	Current State	Recommended Improvement
Site24x7 token storage	Hardcoded in script	Move to environment variable or .env file
InfluxDB token storage	Hardcoded in script	Move to environment variable or .env file
Email password storage	Hardcoded in script	Move to environment variable or .env file
API access level	Read-only on Site24x7	Correct — never use admin credentials for automation
Data in transit	HTTPS for all API calls	Already secure — no change needed
Git safety	Script may be committed	Add .env to .gitignore before any Git commit
Token rotation	No rotation policy	Regenerate tokens every 6 months

10.1 Moving to Environment Variables (Recommended)

```
# Create a file named .env in the same folder as main.py
# Add these lines to .env:
SITE24X7_TOKEN=your_actual_token
INFLUX_TOKEN=your_influx_token
EMAIL_PASSWORD=your_email_password

# In main.py replace hardcoded values with:
from dotenv import load_dotenv
import os
load_dotenv()
SITE24X7_TOKEN = os.getenv("SITE24X7_TOKEN")
INFLUX_TOKEN   = os.getenv("INFLUX_TOKEN")
EMAIL_PASSWORD = os.getenv("EMAIL_PASSWORD")

# Install: pip install python-dotenv
# Add to .gitignore: .env
```

11. Grafana Migration Context

This automation project exists alongside the broader initiative to fully migrate monitoring from Site24x7 to a Prometheus and Grafana stack. The Python suite acts as a bridge during the transition.

Phase	Tool	Status	Notes
Current	Site24x7	Active	Python script pulls data and generates reports
In Progress	Grafana dashboards	In Progress	Building dashboards to replace Site24x7 UI views
Planned	InfluxDB	Setup phase	Time-series backend for Grafana data source
Future	Prometheus	Planned	Replace Site24x7 monitoring entirely
End state	Prometheus + Grafana	Target	Python script becomes redundant for live monitoring

12. Version History

Version	Date	Changes	Author
1.0	Earlier	Initial — Site24x7 API + Excel + basic email	DevOps Team
2.0	11 August 2026	Added InfluxDB push, Grafana screenshot, HTML email with summary boxes, auth validation guide	DevOps Team

13. References

- Site24x7 API Docs: <https://www.site24x7.com/help/api/>
- Zoho Auth Token: <https://accounts.zoho.in/apiauthtoken/create>
- InfluxDB Download: <https://portal.influxdata.com/downloads/>
- InfluxDB Python Client: <https://github.com/influxdata/influxdb-client-python>
- Grafana Download: <https://grafana.com/grafana/download>
- Playwright Python: <https://playwright.dev/python/>
- Outlook App Passwords: <https://account.microsoft.com/security>
- Python dotenv: <https://pypi.org/project/python-dotenv/>